Durham University

Examination Paper

Examination Session:	Year:	Exam Code:
May/June	2021	ENGI3451-WE01

Title: **MEng/BEng:** Electronics and Communications 3		
Time Allowed:	2 hours	
Additional Material provided:	No	
Materials Permitted:		
Calculators Permitted:	Yes	Models Permitted: Those from the Casio fx-83 and fx-85 series.
Visiting Students may use dictionaries: Yes		

Instructions to Candidates:	Answer ALL questions. All relevant workings must be shown.

Revision:	

Section A

Question 1

(a) Consider the following double-sideband suppressed-carrier (DSB-SC) amplitude modulated signal

$$s(t) = 10\left[2\cos(600t) + 4\sin(10^3\pi t)\right]\cos(16 \times 10^5\pi t).$$

(i) Determine the power content of $s(t)$.

(ii) Determine the spectrum and bandwidth of $s(t)$.

[40%]

(b) Consider a discrete memoryless source that produces outputs x_1, x_2, x_3, x_4, x_5, x_6 and x_7 with probabilities 0.21, 0.16, 0.1, 0.08, 0.3, 0.1 and 0.05, respectively.

(i) Design a Huffman code for this source over an alphabet of size $D = 2$ (binary) and determine its average codeword length.

(ii) Determine the entropy of this source. Comment on the result.

[45%]

(c) Consider a (15,11) Hamming code.

(i) Find the parity check matrix of this code in the systematic form.

[15%]

Question 2

(a) Consider the following frequency modulated signal

$$s_{FM}(t) = 20\cos\left(2\pi 10^8 t + \sin\left(2\pi 10^4 t\right) + 2\sin\left(8\pi 10^4 t\right)\right).$$

Assume that the frequency sensitivity factor is $k_f = 10$ kHz/V.

(i) Determine the expression of the message signal.

(ii) What is the bandwidth of $s_{FM}(t)$?

[30%]

(b) A discrete memoryless source produces outputs x_1, x_2, x_3, x_4 and x_5 with probabilities $\frac{1}{3}$, $\frac{1}{9}$, $\frac{1}{3}$, $\frac{1}{9}$ and $\frac{1}{9}$, respectively. Assume coding over an alphabet of size $D = 3$ (ternary). Note that a ternary code is one which uses three alphabets in coding, e.g., $\{0, 1, 2\}$.

(i) Design a Shannon code for this source and determine its average codeword length.

(ii) Determine the entropy of this source. Comment on the result.

[45%]

(c) The generator matrix for a linear binary code is

$$\mathbf{G} = \begin{bmatrix} 1 & 0 & 0 & 1 & 1 & 0 \\ 0 & 1 & 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 & 1 & 1 \end{bmatrix}.$$

(i) Find all the codewords of this code.

(ii) If the received sequence (using hard decision decoding) is $\mathbf{y} = 011010$, what is the transmitted sequence using minimum-distance decoding?

[25%]

Section B

Question 3

The MOSFETs in the circuit shown in Fig. Q3 have the following parameters:
$\mu_n C_{OX} = 180\ \mu\text{A/V}^2$
$\mu_p C_{OX} = 60\ \mu\text{A/V}^2$
$V_T = 0.5$ V
The aspect ratios W/L of the MOSFETs' channels are shown in the circuit. Assume that the channel modulation effect is negligible. Additionally, $V_{DD} = 2$ V and $C_0 = 1$ nF.

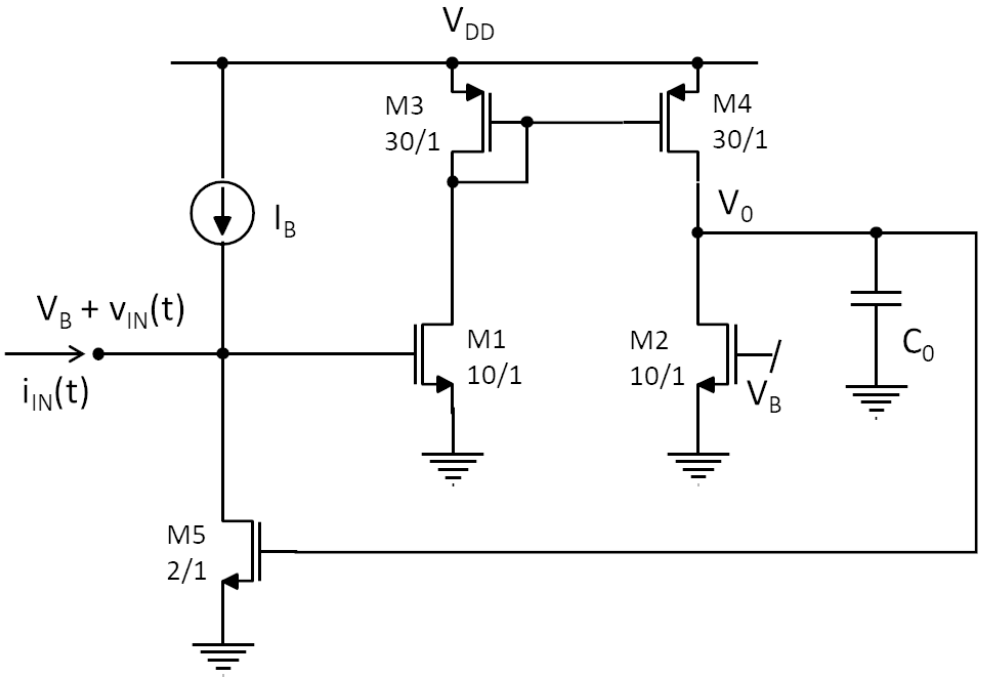


Figure Q3: The circuit for Q3.

(a) Find an expression for I_B as a function of V_B which ensures that $i_{IN} = 0$ A when $v_{IN} = 0$ V. *Hint*: begin with determining the gate voltage of M5. **[20%]**

(b) Assuming sinusoidal signals in the complex domain ($s = i\omega$), derive an expression for the input impedance $Z_{IN}(s) = V_{IN}(s)/I_{IN}(s)$. *Hint*: start by finding the transfer function between v_{IN} and V_0. Use the small-signal approximation throughout. **[30%]**

(c) Assuming that $V_B = 0.8$ V, sketch a plot of the real and imaginary parts of the input impedance Z_{IN} as a function of the frequency. **[20%]**

continued

(d) Give an example of a possible application of this circuit. [**15%**]

(e) Comment on the effect of a non-negligible channel modulation coefficient on the input impedance Z_{IN}. [**15%**]

continued

Question 4

(a) Consider the output characteristic (i.e. V_{OUT} vs. V_{IN}) of a symmetric CMOS inverter. Write an expression for V_{OUT} as a function of V_{IN} for the different operating regions. Neglect the channel-modulation effect, and the region of the output characteristic where both MOSFETs are in saturation. Assume the same threshold voltage (in magnitude) for both MOSFETs.

[30%]

(b) A simple error-correcting method for transmitting digital signals consists of sending the same bit three times (known as a (3-1) repetition code). Democratic voting determines the value of the detected bit: for example 111 is an error-free 1, 101 is detected as a 1 (with one error) as the majority of bits is 1. The truth table of this detector is:

A	B	C	OUT
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	1
1	1	1	1

where A, B, and C are the inputs, and OUT is the output.

(i) Draw a symmetric CMOS circuit which realises this detector.

(ii) Determine the gate size of all MOSFETs, assuming that the minimum channel length is L_{MIN} =32 nm, the minimum channel width is W_{MIN} = 64 nm, and that $\mu_n/\mu_p = 4$.

[40%]

(c) Assume that the input of the detector in (b) changes at a frequency f_0 = 100 MHz and that only error-free codes occour (i.e. either ABC = 111 or ABC = 000) and they appear with equal probability. Determine the average dynamic power dissipation of the detector assuming that V_{DD} = 1.5 V, the MOSFETs gate capacitance is approximated by $C_G = W \times L \times C_{OX}$ where C_{OX} = 0.5 μF/cm^2, and the load capacitance at the output is C_L = 1 fF. **[30%]**

END